

K-Nearest Neighbors (kNN)

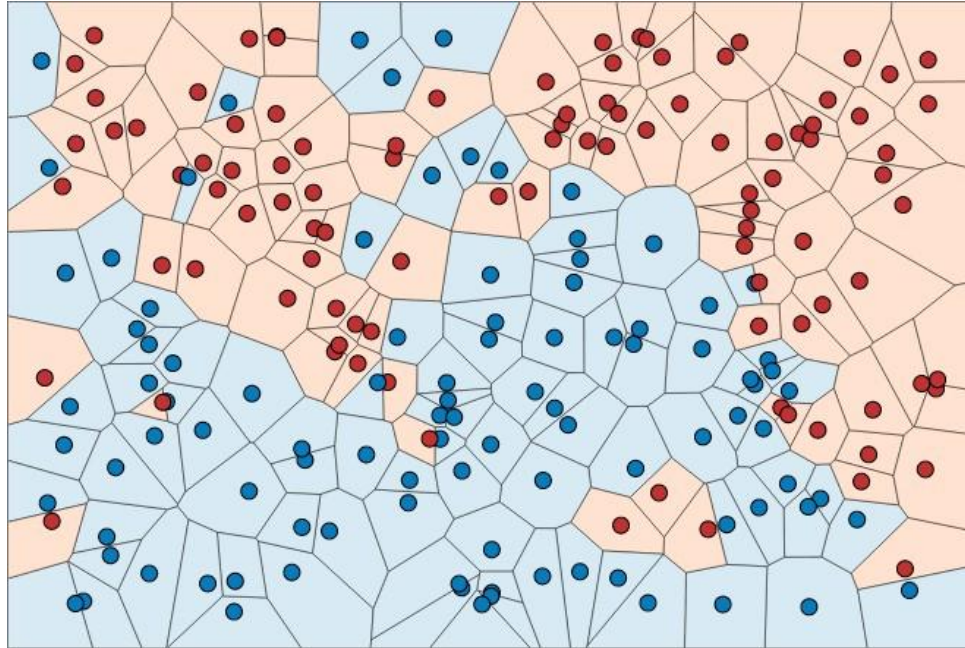


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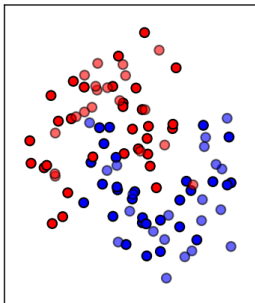
Classification Task

- Classification is probably the most common data mining task
- A supervised method that includes two or more classes for the categorical target variable
- Examples
 - Banking: determine whether a mortgage application is good or not
 - Advertising: determine whether a customer is going to click on the ad or not
 - Medicine: diagnose whether a disease is present or not present
 - Law: determine if a signature is fraudulent or not
 - Security: identify whether a certain financial transaction represents a threat
 - ...

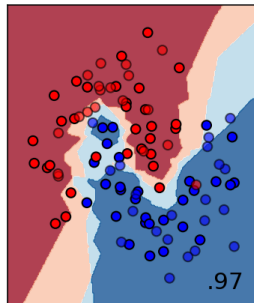
Regression vs. Classification

- Basic difference
 - In classification, we have dependent variables that are categorical.
 - In regression, we have dependent variables that are continuous values.
- The classification problem is all about drawing the “right” decision boundary.
- Approaches to classification
 - **Nearest-neighbor**
 - Naïve Bayes
 - Decision trees / Random Forest
 - Logistic regression
 - Artificial neural networks
 - Linear Support Vector Machine (SVM)
 - ...

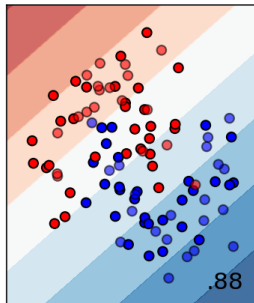
Input data



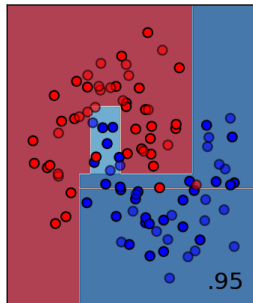
Nearest Neighbors



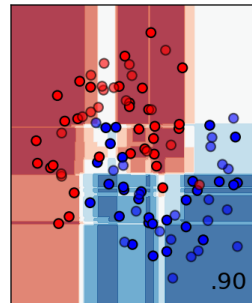
Linear SVM



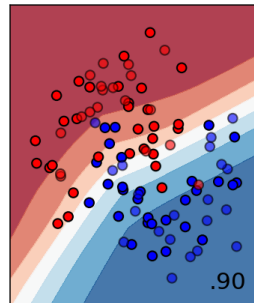
Decision Tree



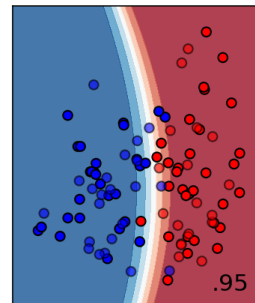
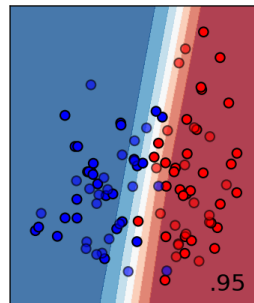
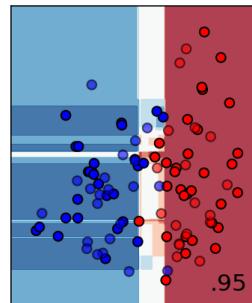
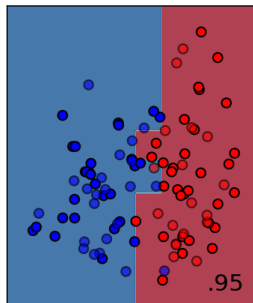
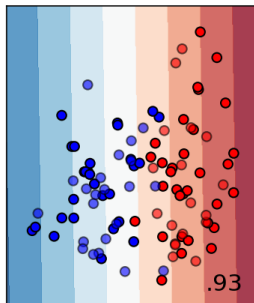
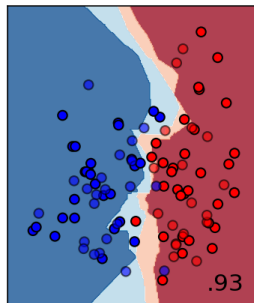
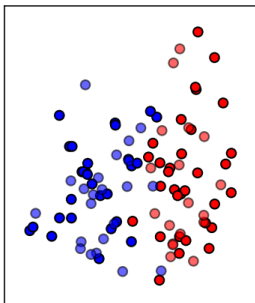
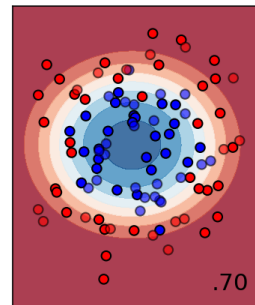
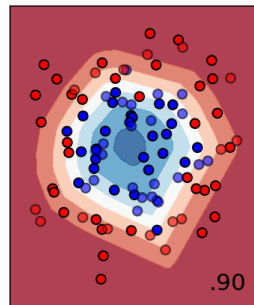
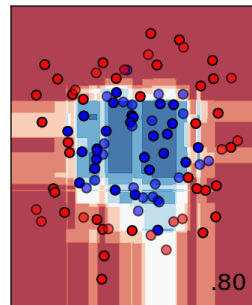
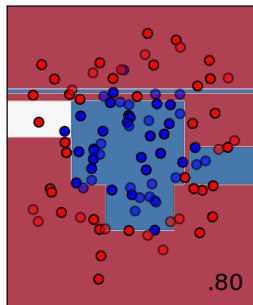
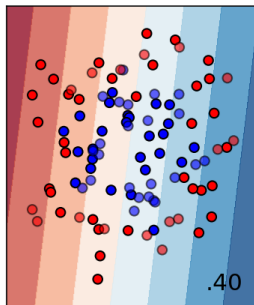
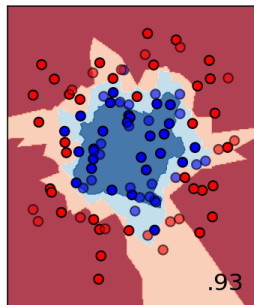
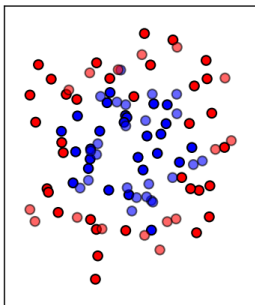
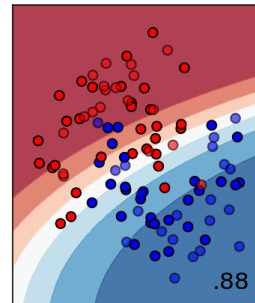
Random Forest



Neural Net



Naive Bayes

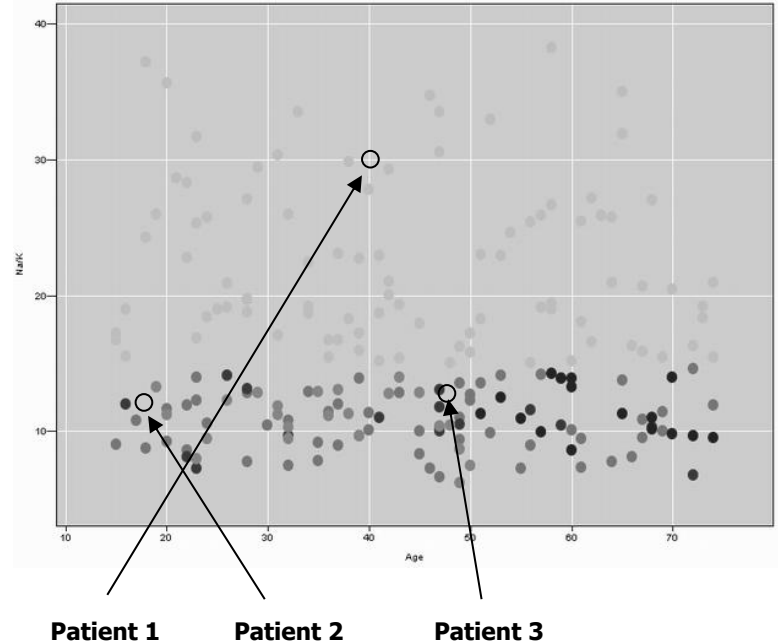


k-Nearest Neighbor Algorithm

- Simple idea: classify a record like other “similar” records
 - For a given record to be classified, identify ‘k’ ‘nearby’ records, where ‘k’ is up to the analyst
 - “Near” implies some measure of distance... intuitively, we are looking for records with similar predictor values $X_1, X_2, \dots X_p$
 - Classify the record as whatever the predominant class is among the nearby records (the “neighbors”)
 - Usually, “**majority vote**” determines class
- [Demo](#)

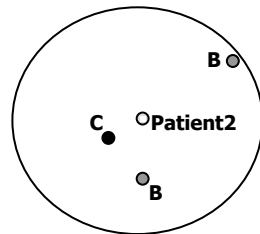
k-Nearest Neighbor: An Example

- Classifying the type of drug 3 patients should be prescribed
- The training set consists of 200 patients with sodium-potassium Na/K ratio, age, and drug attributes
- A “drug” overlay is shown where Light points = drug A, Medium points = B, and Dark points = drug C
- This scatter plot of Na/K against Age shows the records in the training set that patients 1, 2, and 3 are most similar to
- Which drug should Patient 1 (40-years-old with a Na/K ratio of 30) be prescribed?
 - Since Patient 1's profile places them in the scatter plot near patients prescribed drug A, we classify Patient 1 as drug A
 - All points near Patient 1 are prescribed drug A, making this a straightforward classification



k-Nearest Neighbor: An Example - Patient 2

- Patient 2 is 12 years old with a Na/K ratio = 18. A close-up shows the neighborhood of training points in close proximity to Patient 2
- Suppose we let $k = 1$ for our k-Nearest Neighbor algorithm
 - This means we classify Patient 2 according to whichever single point in the training set it is closest to
 - In this case, Patient 2 is closest to the Dark point, and therefore we classify them as drug C
- Suppose we let $k = 2$ and reclassify Patient 2 using k-Nearest Neighbor
 - Now, Patient 2 is closest to a Dark point and Medium point
 - How does the algorithm decide which drug to prescribe?
 - A simple voting scheme does not help -> use distance weights to break ties
- However, with $k = 3$, voting determines that two of the three closest points to Patient 2 are Medium
 - Therefore, Patient 2 is classified as drug B
 - Note that the classification of Patient 2 differed based on the value chosen for k



Considerations when using k-Nearest Neighbor

- How is the distance between points measured?
 - The one we've used (intuitively) is called Euclidean distance
 - Remember normalization/standardization... ? When distances can vary greatly between different columns, it's a good idea to standardize your data beforehand
- How is the information from two or more neighbors combined when making a decision?
 - Should all points be weighted equally, or should some points have more influence? (next slides)
- How many neighbors k should be used? (next slides)

How to combine the most similar k records?

- Simple Unweighted Voting

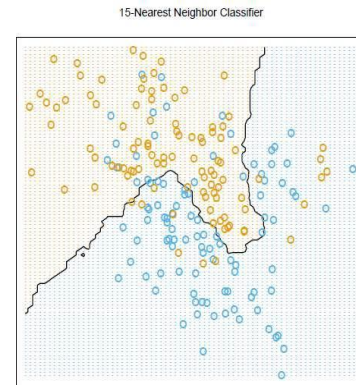
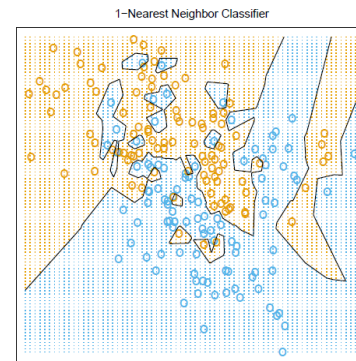
- This is the most simple combination function
- Decide on the value for k to determine the number of similar records that “vote”
- Compare each unclassified record to its k nearest (most similar) neighbors according to the Euclidean Distance function
- Each of the k similar records vote

- Weighted Voting

- The analyst may choose to apply weighted voting, where closer neighbors have a larger voice in the classification decision than do more distant neighbors
- In weighted voting, the influence of a particular record is inversely proportional to the distance of the record from the new record to be classified

How to choose k?

- k is the number of nearby neighbors to be used to classify the new record
 - k=1 means use the single nearest record
 - k=5 means use the 5 nearest records
- Typically choose that value of k which has lowest error rate in test data
- Low values of k (1, 3, ...) capture local structure in data (but also noise)
- High values of k provide more smoothing, less noise, but may miss local structure
- The extreme case of $k = n$ (i.e., the entire data set) is the same as the “naïve rule” (classify all records according to majority class)



Summary

- Advantages

- Simple, no assumptions required
- Handles nonlinear decision boundaries. Effective at capturing complex interactions among variables without having to define a statistical model

- Shortcomings

- There is no model, not interpretable
- Requires some (random) tinkering with parameters
- Can fail for very large datasets due to curse of dimensionality
 - Expected distance to nearest neighbor increases with dimensions (with many dimensions, all records end up “far away” from each other)
 - Try reducing dimension of predictors (e.g., with PCA)
 - For very large datasets: use cosine similarity (cosine distance) instead.
- Can be slow for very large datasets
 - Not memory efficient, requires storage of the training data